

Info. No.: 341402		Item Name: Bearing Wear Monitoring			Item Id.: 3091686-8	
Scale:	Size: A4	Product Type: Specification of Signal Processing and Operational Limits for Engines Coupled to a Fixed Pitch Propeller			Projection: 	Page No.: 1 (38)
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20080208	HWC	HWC		Moved to template and draft archived in Documentum		00
20120927	MTO	HWC	X3	Improved Learning procedure and Rapid Wear		05
20171127	NIHE	HWC	X3	General revision with changes in learning procedure, Rapid Wear algorithm, alarm limits, data storage, engine control system interface		06
20180918	NIHE	HWC	X4	Small modification in the treatment of rapid wear during learning.		07

Introduction

Bearing Wear Monitoring (BWM) is a method of monitoring the condition of the three major bearings of a two stroke crosshead engine. These are main-, crank pin-, and crosshead bearings. The method is based on the fact that any change in bearing wall thickness in the loaded part of one of these bearings (due to wear or seizure) will result in a corresponding change of the Bottom Dead Centre (BDC) level of one or more of the crossheads relative to the engine structure. In case of crank pin- or crosshead bearing wear, the BDC level of the cylinder in question will change, while in case of main bearing wear, the BDC level of the two cylinders adjacent to the worn bearing will change.

The primary aim of BWM is to detect a beginning bearing failure before it develops to an extent where heat is causing damage to other parts than the bearing shell. These could be the crosshead pin, shaft journal, or bearing housing. Such damage will generally be the result if the bearing lining (white metal or tin-aluminium) is worn through and contact between the shaft and the bearing shell steel backing occurs.

There are two different methods to analyse the BDC level signals. One method analyses sudden changes in the BDC level while the other method analyses if there is any slow wear in the bearings. Both methods are described in this document.

Bearing wear monitoring systems are controlled components, level 2. Before starting up of production a first time approval must be performed in accordance with MAN B&W Quality Control No. 0743665-8; Bearing Wear Monitoring System. First-Time Approval.

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Scope

This document specifies the signal processing to be performed in order to obtain values that can be compared to operational limits such as alarm limits or slowdown limits. It is assumed that a true signal representing the actual distance from each guide shoe in its BDC to a sensor (two for each cylinder) is available for every passage of BDC. The signal is to be measured with a repeatability of +/- 0.01 mm. The technical means of obtaining the signal is outside the scope of this document.

Major changes in Revision 6

- Section 4.3: Definition of Δ RPM value is changed.
- Section 2.1: Construction of the rough compensation curve is changed.
- Section 2.3: Construction of the final compensation curve is changed.
- Section 4: Major changes to the rapid wear algorithms.
- Section 5: Three readjustment procedures have been added replacing the one procedure stated in earlier document revisions.
- Section 6: Major changes to the requirements for data storage.
- Section 7.2: Rapid wear alarm limits are changed.

Major changes in Revision 7

- Section 4.5: Changes to the reset procedure at updates of the learning curves.
- Section 5.2.4: Changes to the rapid wear reset procedure when making sensor adjustments.
- Section 6.2: Small change to the specification of storage.

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1 Overview of System Responses

The BDC levels are recorded, both fore and aft, at every revolution of the engine by proximity sensors. The sensors are attached to the guide bars and directed towards the lower edge of the guide shoes. In case of wear in the loaded part of a bearing, or in case of bearing seizure, a comparison with the normal BDC level of the cylinder(s) in question will show abnormal values. Depending on the failure mode detected the BWM-system is to deliver signals to the engine control system requesting a slowdown, an alarm or a pre-warning. The interface signals are described in chapter 8.

The BDC level strongly depends on the engine speed because elastic deformation of the engine structure varies with the engine speed. So the normal BDC level mentioned above is speed dependent. It is to be established during a learning period. The learning mode has several steps designed to gradually increase the accuracy of the system. The first rough compensation curve is quickly made allowing for early protection from bearing failure. The learning mode and the different steps in creating the final compensation curve defining the normal BDC level is described in chapter 2.

The signals in the system are to be measured with a repeatability of ± 0.01 mm. The signals from the sensors will contain some variation. There will be a variation of the guide shoe BDC level due to variations in cylinder firing pressure, variations in engine temperature, waves of the sea and hull deformations from variations in vessel draft. Some of these variations are compensated for by various treatments of the measured data.

The BWM-system is to contain two separate sets of algorithms designed to detect slowly developing wear and rapidly developing damages respectively.

The sequence of processes for analysing slow ongoing wear is described in chapter 3. The sequence of processes for analysing rapidly developing damages is described in chapter 4.

When the various deviations from reference values have been calculated they are compared with corresponding warning and alarm levels which are given in chapter 7. If the magnitude of a deviation has passed a limit, this has to be stored in a log file, and necessary actions have to be taken. Chapter 6 describes the data to be stored.

If sensors are changed or relocated the system no longer displays the correct values of wear. Chapter 5 describes various procedures which are to be available to the operator of the BWM-system to correct for such issues.

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2 Speed (RPM) Compensation

The engine speed significantly influences the BDC level. A speed variation of more than 0.3 mm is possible, depending on engine size. A compensation of this influence is necessary. The influence is individual for each installation. Among other things, it depends on the engine layout and the alignment of the shaft line. Therefore, the compensation must be established individually for each sensor for each installation.

The speed compensation is to be made based on a look-up table containing the normal rpm-dependent signal values designated reference values. Since this normal values may change significantly within small changes in engine rpm, the look-up table must have a fineness corresponding to dividing the speed range between 20% and 110% of nominal engine speed¹ into no less than 100 sub-ranges, in the following designated "speed points".

The range below 20% of nominal speed is considered irrelevant for BWM. Likewise, astern running and running within the torsional vibration defined barred speed range is not to be covered.

The look-up table of engine speed compensation must be established during a "learning process" performed by the BWM system itself. The learning process will last for 500 running hours during which the system is in learning mode establishing a full speed compensating look-up table for each sensor.

It is important that the system can offer protection of the bearings from the earliest possible time. Therefore, a rough compensation curve is to be established, as soon as a 10 minute average value from one fixed engine speed is achieved. This curve is then re-adjusted so three fixed engine speeds are used to start with during the learning process.

As soon as the first speed point has been established the system can provide bearing wear protection.

2.1 Rough Compensation Curve

During the learning mode, the system initially uses a rough compensation curve based on 3 fixed speeds for 10 minutes at each speed. The three fixed speeds must be spaced by minimum 20% of nominal speed and located in the following ranges:

¹Throughout this document the term nominal engine speed is used extensively. Nominal engine speed refers to the MCR (Maximum Continuous Rating) speed of the engine. E.g. in an S50ME-C8 engine the nominal engine speed for an L1 rated engine is 127 rpm while it is 102 rpm for an L3 rated engine.

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1. 20-50% of nominal speed
2. 50-80% of nominal speed
3. 80-100% of nominal speed

The order of running the three speed points is unimportant.

Fixed speed is defined as fixed engine speed $\pm 2\%$ of nominal engine speed.

During each of the three periods, signals of speed and sensor values (distance) are sampled and average speed and sensor values are calculated for each speed.

After recording data for 10 minutes of the first speed the first rough compensation curve (see figure 2.1) for each sensor is generated, based on this point and a slope corresponding to a certain change in sensor distance between 20% and 110% engine speed. See chapter 7.

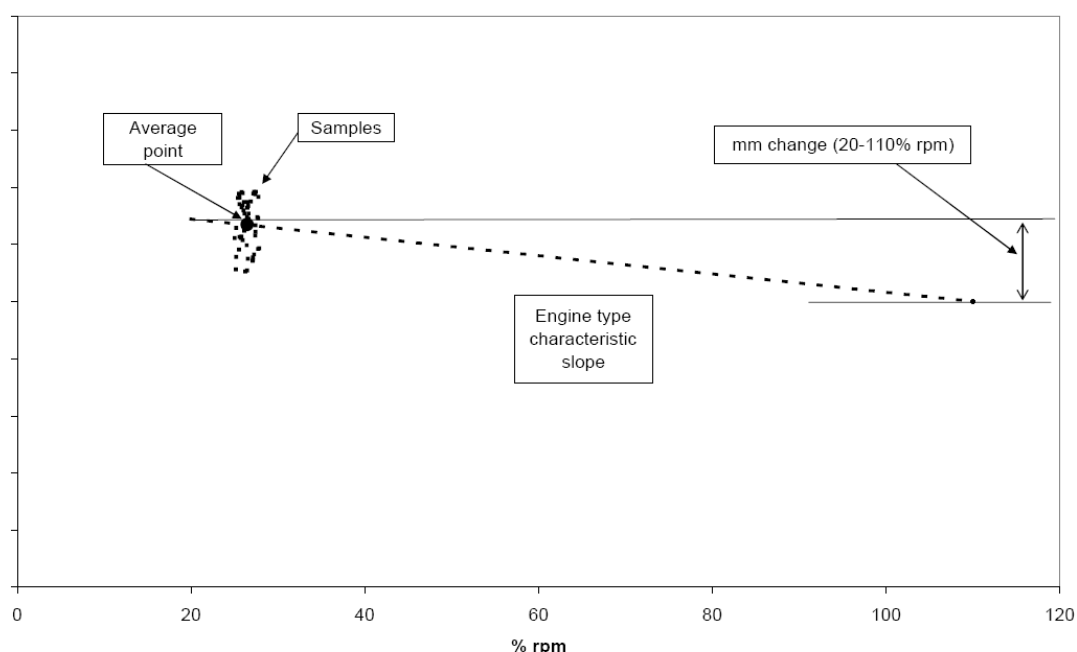


Figure 2.1, Preliminary rough compensation curve

When the first rough compensation curves have been created the rapid wear values are reset.

$$\text{Rapid Slow} = \text{Rapid Fast} = 0$$

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This first compensation curve serves as reference until the second speed has been covered. Then a new compensation curve is generated between 20% and 110% engine speed, based on the slope between the two covered speed points. For the second point also 10 minutes of recorded data is used.

When the compensation curves have been modified the rapid fast and rapid slow values are reset according to the method defined in chapter 4.5. Such resets are to be performed every time the compensation curves are modified.

When sampling at the 3 speeds has been completed, the rough compensation curve for each sensor is built from these 3 points by interpolation and extrapolation to cover the speed range 20-110%. See figure. 2.2.

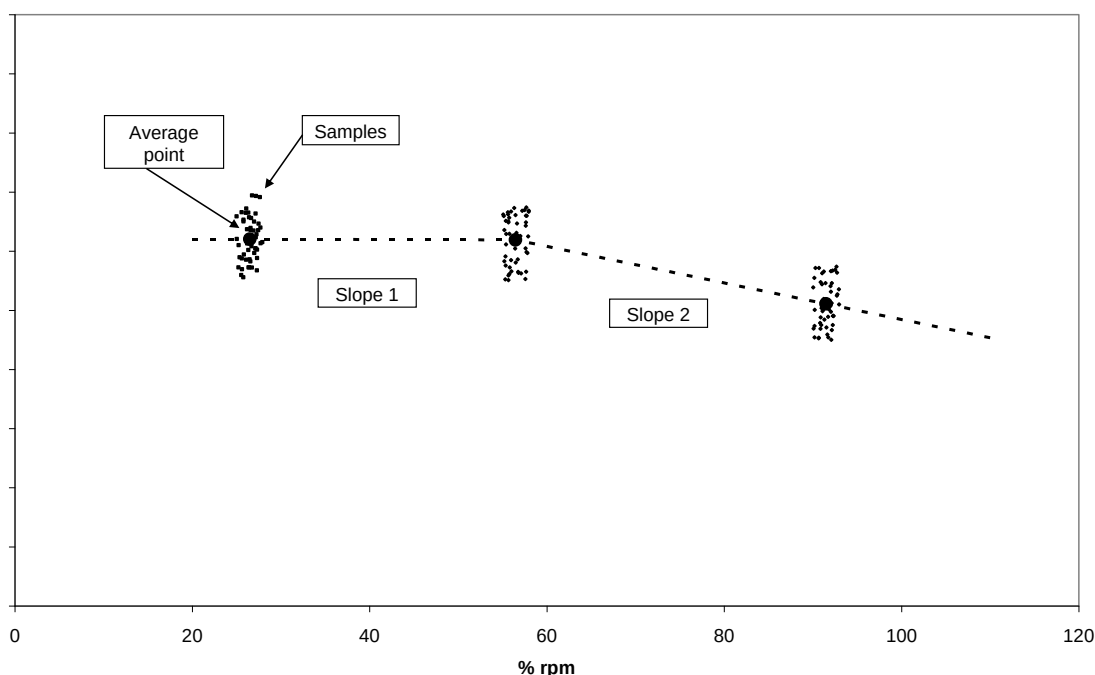


Figure 2.2, Rough compensation curve

During the creation of the rough compensation curves elevated rapid wear alarm levels and elevated slow wear alarm levels are used. See chapter 7.

2.2 Refining the Compensation Curve

With the rough compensation curves established, the system will proceed into the next step of the learning mode. At this stage, the engine is operated freely. In this mode, the system is sampling sensor values in relation to the, say, 100 speed points. Once

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1000 samples (corresponding to 1000 engine revolutions) have been acquired within one speed point, an average is calculated, defining the reference value.

A sample is only considered valid if the engine is running forward and steadily. If the change in engine speed from one sample to the next is higher than 5 % of the nominal engine speed the engine is considered running unsteadily and both samples are to be discarded.

Each time a new valid point has been achieved, the speed compensation curves are to be updated by interpolating between the valid points. See figure 2.3.

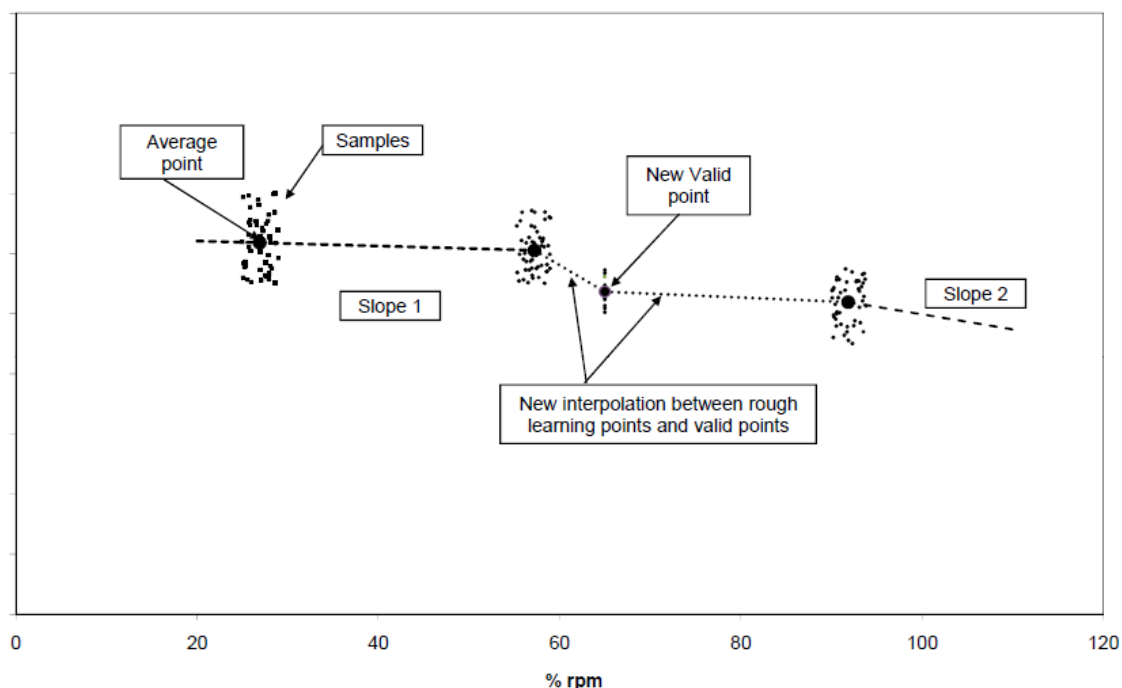


Figure 2.3, Updated compensation curve

The three points established for the rough compensation curves are to be seen as valid until a new speed point has been achieved within ± 2 % of nominal engine speed of the original speed point. See figure 2.4.

Speed points for higher speeds than the highest valid speed point or lower than the lowest valid speed point are extrapolated using the slopes from the rough compensation (slope 1 and slope 2).

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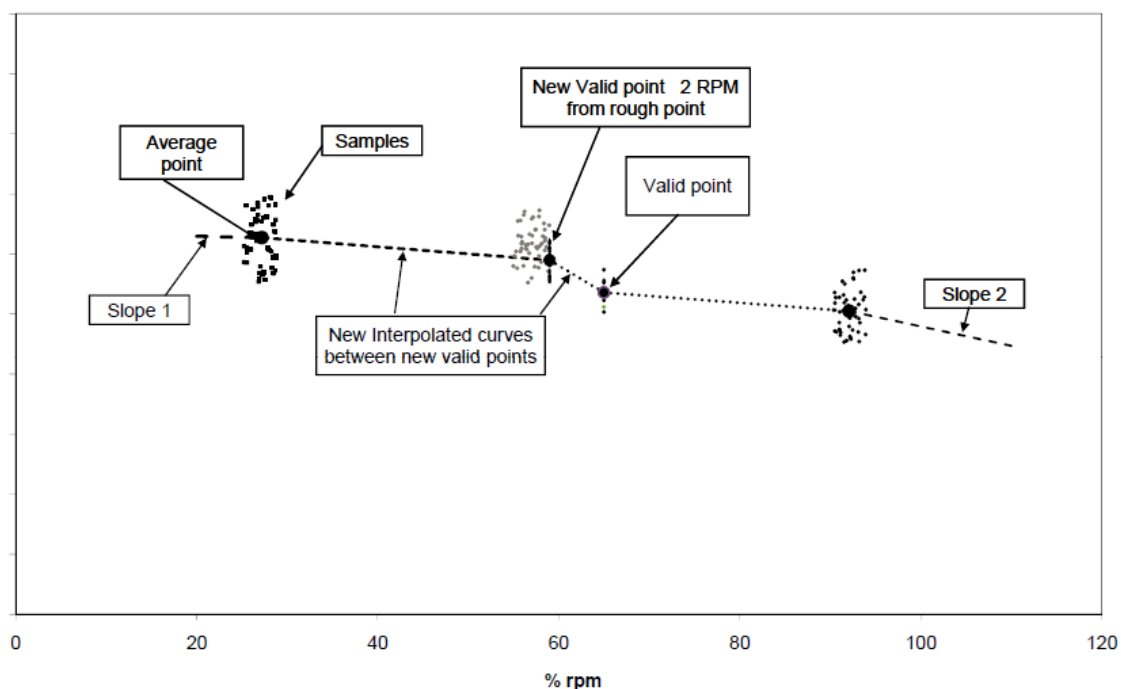


Figure 2.4, Updated compensation curve with updated initial speed point

The sampling of reference values is to continue for 500 hours of engine operation. In this period, the average at each valid speed point is re-calculated every 50 hours based on all available samples apart from the original fixed 3 points, and the look-up table is updated accordingly. Only speed points with 1000 or more samples are to be updated. If the reference value at any speed point upon an update is changed by 0.2 mm or more when compared to the first obtained valid value (based on 1000 samples), an alarm has to be released, and the system must indicate this situation.

When a speed compensation curve update is made, the rapid wear values are to be reset. See chapter 4.5.

During the 500 hours of refinement of the compensation curves elevated rapid wear alarm levels and elevated slow wear alarm levels are used. See chapter 7.

2.3 Final Compensation Curve

When finishing the 500 hours of refinement, reference values at some speed points may not have been established due to lack of samples (less than 1000) taken at this speed. Such reference points are now defined based on a weighted average between the average value ($Value_{Avg, Nhits}$) of the available samples and the value ($Value_{interp}$) interpolated from the nearest two valid speed points. If the speed point is located in

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the beginning or end of the curve where extrapolation is used $Value_{interp}$ refers to the extrapolated value.

$$REF = 1/1000 * (N_{hits} * Value_{Avg, N_{hits}} + (1000 - N_{hits}) * Value_{interp}) , \text{ if } N_{hits} < 1000$$

If, after defining these reference points some speed points are still not defined because no samples were taken the remaining reference points are defined by interpolating between the nearest points with defined reference points, or - if necessary - by extrapolating according to the slopes calculated according to calculation of slopes during learning mode. See figure 2.5.

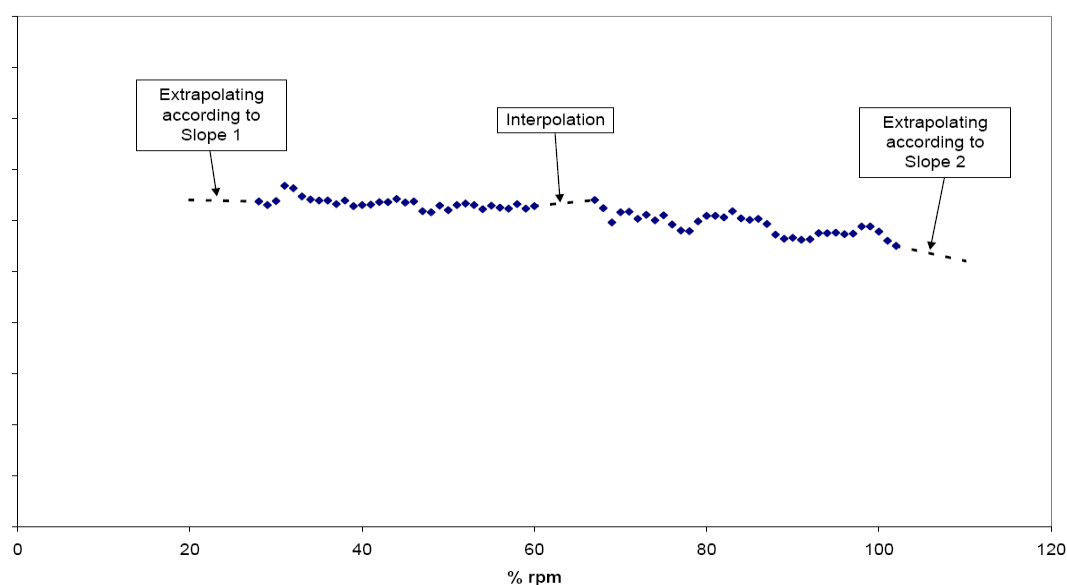


Figure 2.5, Final compensation curve

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3 Slow Wear Monitoring

Slow wear is characterised by slowly decreasing bearing lining material thickness. The bearings in MAN B&W 2-stroke diesel engines usually do not experience significant wear of the lining materials during the lifetime of the engine. I.e. slow wear is unusual but can be caused by solid to solid contact during turning or due to lack of lubrication oil. Or it could be caused by an erosive process.

The slow wear algorithm compares the present BDC-levels with the initial levels defined by the compensation curves.

Values achieved within the torsional vibration defined barred speed range are to be disregarded.

3.1 Noise Removal

To reduce the influence of noise, low pass filtering of the speed compensated sensor values (S_{comp}) is to be applied. The filter establishes a moving average which updates at every revolution when a new sensor value S_N is received. The "update ratio" is to be 5%. Thus, the filtered sensor values (SF) are calculated:

$$S_{comp} = S_N - S_{ref}$$

$$SF_{new} = SF_{old} * (1-x) + S_{comp} * x, \quad x=0.05$$

3.2 Deviation Calculations

Some variations in the engine condition may influence all the sensor signals in the same way. As an example, a general change in engine temperature will cause all the signals to change by almost identical values. In order to exclude the common change of BDC level, a calculation of the difference between the individual filtered sensor signal and the average of the remaining signals is to be done. As an example, if 8 filtered sensor signals ($s_1 - s_8$, corresponding to a 4 cylinder engine) are available, the deviation of sensor no. 5 is calculated as

$$d(s_5) = s_5 - (s_1 + s_2 + s_3 + s_4 + s_6 + s_7 + s_8)/7$$

-and designated sensor deviation.

Experience has shown that sensors on fore and aft side of the same cylinder have a tendency to produce variations in counter phase: one signal increases while the other signal decreases. Such variation may be caused by variation in thrust causing a

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longitudinal shear formed deformation of the engine structure, or may be induced by hull deformation. It can be eliminated by calculating the average of the two signals originating from one cylinder. Using the above example of a 4 cylinder engine, the deviation of cylinder 3 (where sensors 5 and 6 are mounted) is calculated as

$$d(c_3) = (s_5 + s_6)/2 - (s_1 + s_2 + s_3 + s_4 + s_7 + s_8)/6$$

-and designated cylinder deviation.

By calculating the cylinder deviation, the signal scatter is reduced compared to sensor deviation.

3.3 Evaluation of Data

Different bearing wear scenarios will cause sensor value, sensor deviation or cylinder deviation to give the clearest reaction. Therefore, all of these three variables are to be evaluated.

Sensor deviation will give a clearer reaction than cylinder deviation if a main bearing is wearing.

Cylinder deviation will give a clearer reaction than sensor deviation if a crosshead bearing is wearing.

Sensor value will give the clearest reaction if all bearings are deteriorating in a common erosive process.

3.3.1 Alarm and slowdown

The filtered sensor values and calculated sensor- and cylinder deviations are to be compared with the corresponding levels for alarm and slowdown. The evaluation of acquired data has to be done at least every 30 revolutions.

If an alarm limit is exceeded the system is to send an alarm or slowdown request to the engine control system. Furthermore, the system is to store data in frozen copy. The details of the data to store are given in chapter 6.

All alarm levels and limits are stated in chapter 7.

3.3.2 Pre-warning

The system is to report a "Pre-warning" status of a cylinder unit if the six-hour average "compensated sensor value" of one (or both) of the sensors in this cylinder changes

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by more than the Pre-warning level. See chapter 7. The references are the obtained values from the learning process.

The pre-warning is to indicate that the condition of any monitored bearings has started to change. If a pre-warning occurs, no immediate actions have to be taken, as long as an alarm or slowdown limit has not been reached. By acknowledging the pre-warning, the pre-warning limit for the actual sensor is to be increased by 0.25 mm. Any pre-warning and any reset of the pre-warning reference is to be reported to the log file (see chapter 7).

3.4 Flow Chart for Slow Wear Detection

A flow chart illustrating the various steps in slow wear detection is shown in figure 3.1.

A new set of sensor values are available at every revolution of the engine. The evaluation may be performed at intervals of up to 30 engine revolutions. This means that the first four steps in the flow diagram must be done at each revolution, while the last steps must be done at least once every 30 engine revolutions.

Values achieved within the torsional vibration defined barred speed range are to be disregarded.

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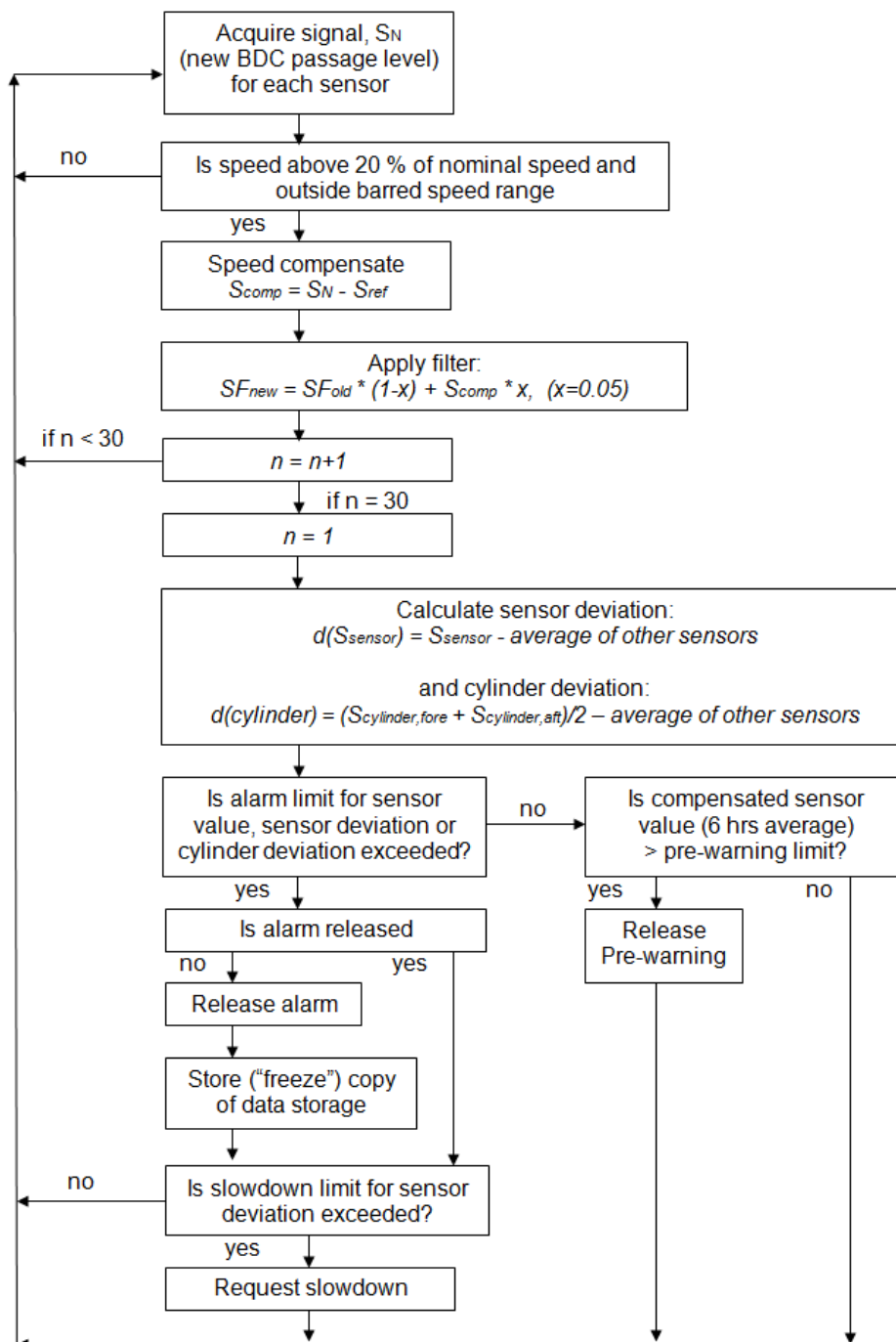


Figure 3.1, Flow chart for slow wear evaluation.

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4 Rapid Wear Monitoring

Rapidly developing bearing failures can be caused by e.g. filter breakage. Such failures give different responses than slow wear. The BDC-level will change relatively quickly from a stable level. The BDC-level may become higher or lower depending on the wear mechanism.

The BWM algorithms to detect such situations consist of a comparison between a fast floating average and a slow floating average. The Rapid Wear values for more sensors are combined in various ways.

If rapid wear is detected a serious situation is likely to exist potentially causing damage to the engine structure within tens of minutes. Engine slowdown is therefore to be requested.

Updating of the speed compensation curves will affect the Rapid Wear values. Therefore, separate algorithms are to correct the Rapid Wear values at updates of the speed compensation curves.

Values achieved within the torsional vibration defined barred speed range are to be disregarded.

4.1 Rapid Wear Value Calculation

The Rapid Wear value for each sensor is calculated using exponential moving averaging of the speed compensated values. By using a slow update coefficient, a reference can be created that reacts slowly on sudden irregularities. By subtracting this reference from an exponential moving average with fast update factor, values to detect irregularities can be created.

The exponential moving averages are created with:

$$Avg_{new} = Avg_{old} (1-x) + Value_{new} \cdot x$$

Where $Value_{new}$ are the speed compensated values.

4.1.1 Rapid Slow

The Rapid Slow value is used to state a present normal level and is designed to react slowly. Therefore, the update factor x is set to 0.0001.

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4.1.2 Rapid Fast

The Rapid Fast value is to react fast on changes. The update factor x is set to 0.2.

4.1.3 Rapid Wear Value

The Rapid Wear Value for each sensor (RWS) is defined as:

$$RWS = \text{Rapid Wear Sensor Value} = \text{Rapid Fast} - \text{Rapid Slow}.$$

Similarly, the average Rapid Wear Value for each cylinder is defined as:

$$RWC = \text{Rapid Wear Average Cylinder Value} = (RWS_{\text{Fore Cyl},m} + RWS_{\text{Aft Cyl } m}) / 2$$

Where m designates the cylinder number

And the average Rapid Wear Value for each two cylinders is defined as:

$$\begin{aligned} RWT &= \text{Rapid Wear Average Twin Cylinder Value} \\ &= (RWS_{\text{Fore Cyl},m} + RWS_{\text{Aft Cyl } m} + RWS_{\text{Fore Cyl},m+1} + RWS_{\text{Aft Cyl } m+1}) / 4 \end{aligned}$$

Figure 4.1 shows how to combine the sensors to calculate the RWC values and figure 4.2 shows how to combine the sensors to calculate the RWT values

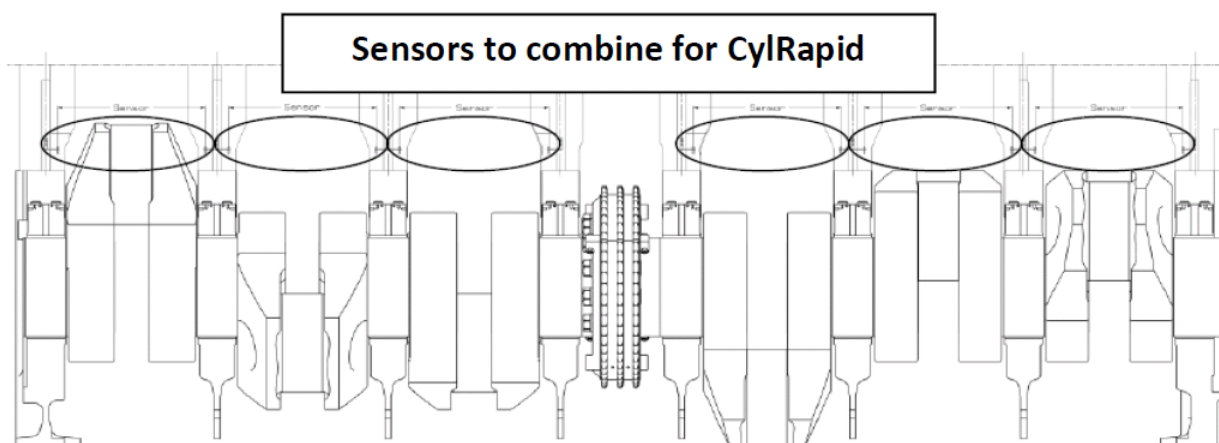


Figure 4.1, Sensors to combine for Rapid Wear Average Cylinder Value.

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4.2 Rapid Wear Deviation Calculations

Some variations in the engine condition may influence all the sensor signals in the same way. As an example, a general change in engine temperature will cause all the signals to change by almost identical values. In order to exclude the common change of BDC level, a calculation of the difference between the individual sensor signal and the average of the remaining signals is to be done. As an example, if 8 sensor signals ($S_1 - S_8$, corresponding to a 4 cylinder engine) are available, the deviation of sensor no. 5 is calculated as

$$dRS_5 = RWS_5 - (RWS_1 + RWS_2 + RWS_3 + RWS_4 + RWS_6 + RWS_7 + RWS_8)/7$$

-and designated Rapid Wear Sensor Deviation.

Similarly, the deviation of the cylinder sum for cylinder 2 in a 4 cylinder engine is calculated as

$$dRC_2 = (RWS_3 + RWS_4)/2 - (RWS_1 + RWS_2 + RWS_5 + RWS_6 + RWS_7 + RWS_8)/6$$

-and designated Rapid Wear Average Cylinder Deviation.

The deviation of the twin cylinder sum for cylinder 2 and 3 in a 4 cylinder engine is calculated as

$$dRT_{2,3} = (RWS_3 + RWS_4 + RWS_5 + RWS_6)/4 - (RWS_1 + RWS_2 + RWS_7 + RWS_8)/4$$

-and designated Rapid Wear Average Twin Cylinder Deviation.

4.3 Combining Rapid Wear Sensor Values

The Rapid Wear values are compared with alarm limits both as single values and combined with neighbour sensors. Three values are calculated: SingleRapid, CylinderRapidAve and TwinCylinderRapidAve.

CylinderRapidAve will usually be the parameter to release an alarm in case of a crosshead bearing or crankpin bearing damage.

Either CylinderRapidAve or TwinCylinderRapidAve will be the usual parameter to release an alarm in case of a crosshead bearing or crankpin bearing damage.

SingleRapid may be the parameter to cause an alarm in case that a sensor is out of order so that CylinderRapidAve or TwinCylinderRapidAve cannot be evaluated.

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4.3.1 SingleRapid

The SingleRapid value for a sensor is defined as the minimum of the absolute values of the Rapid Wear Sensor Value and the Rapid Wear Sensor Deviation.

$$SingleRapid_n = \min(|RWS_n|, |dRS_n|), n = \text{sensor number}$$

4.3.2 CylinderRapidAve

The CylinderRapidAve value for a cylinder is defined as the minimum of the absolute values of the Rapid Wear Average Cylinder Value and the Rapid Wear Average Cylinder Deviation.

$$CylinderRapidAve_m = \min(|RWC_m|, |dRC_m|), m = \text{cylinder number}$$

4.3.3 TwinCylinderRapidAve

The TwinCylinderRapidAve value for two cylinders is defined as the minimum of the absolute values of the Rapid Wear Average Twin Cylinder Value and the Rapid Wear Average Twin Cylinder Deviation.

$$TwinCylinderRapidAve_{m,m+1} = \min(|RWT_{m,m+1}|, |dRT_{m,m+1}|), m = \text{cylinder number}$$

4.4 Dynamic Rapid Alarm limits

There are different alarm levels for SingleRapid, CylinderRapidAve and TwinCylinderRapidAve. The values of CylinderRapidAve and TwinCylinderRapidAve have smaller natural variations than SingleRapid. Therefore, the alarm levels for CylinderRapidAve and TwinCylinderRapidAve are smaller than for SingleRapid.

Increased variations in calculated Rapid Wear are to be expected when the engine speed varies. To account for this, the alarm limits for Rapid Wear are to be dynamic and increased when the engine speed is changed.

The variation of engine speed is defined by:

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$$\Delta RPM_PCT_n = \Delta RPM_n \cdot 100 / RPM_NOM$$

$$\Delta RPM_n = RPM_{high,n} - RPM_{low,n}$$

$$RPM_{low,n} = \min(RPM_{low,n-1}(1-x) + x \cdot RPM_n, RPM_n)$$

$$RPM_{high,n} = \max(RPM_{high,n-1}(1-x) + x \cdot RPM_n, RPM_n)$$

$$x = 0.003$$

For revolutions with ΔRPM higher than 25 % of nominal engine speed (RPM_NOM) the damage monitoring is to be paused and no alarms or slowdowns are to be given. ΔRPM is the only value to be updated when $\Delta RPM > 25$ % of nominal engine speed.

If ΔRPM is between 16 % and 25 % of nominal engine speed, ΔRPM_PCT is to be set to 16 in the following calculations.

The dynamic alarm limit is calculated with the following equation.

$$Alarm_{Dynamic} = Alarm_{Basic} \cdot \left(1 + \frac{\Delta RPM_PCT_n}{k} \right), \text{ where } k=16$$

The basic alarm limits are stated in chapter 7.

4.5 Rapid Wear Sensor Value Reset during Learning Mode

When the compensation curves are changed the rapid fast and rapid slow values are to be corrected to account for the sudden change in reference which could otherwise cause the Rapid Wear values to rise significantly.

The correction is done by calculating an average of speed compensated values from the first 5 revolutions after the change of the compensation curves. Rapid Fast is set to the speed compensation average and Rapid Slow is set to zero. No rapid wear alarms are to be given while the averaging is done.

$$Rapid\ Fast = 5\ Rev\ Speed\ Compensated\ Average$$

$$Rapid\ Slow = 0$$

4.6 Flow Chart for Rapid Wear Detection

A flow chart illustrating the various steps in rapid wear detection is shown in figure 4.2.

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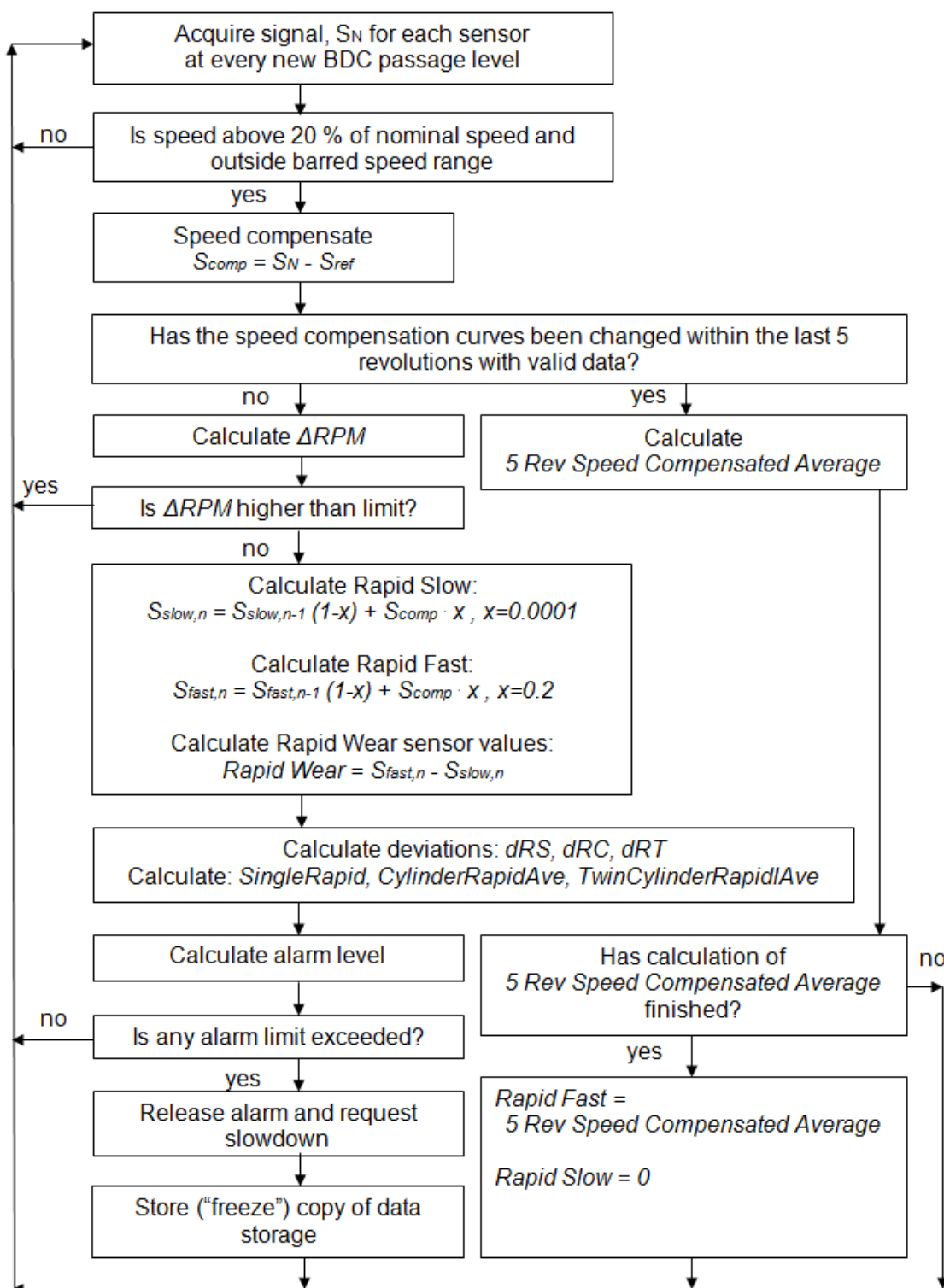


Figure 4.2, Flow chart for rapid wear evaluation.

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5 Sensor Failure

5.1 Corrections for Failed Sensors

In case the signal from one of the distance sensors is lost the calculation of deviations must be corrected accordingly.

E.g. the calculation of sensor deviation for sensor 5 in a 4 cylinder engine

$$d(S_5) = S_5 - (S_1 + S_2 + S_3 + S_4 + S_6 + S_7 + S_8)/7$$

should in case of lost signal from sensor 3 be corrected to

$$d(S_5) = S_5 - (S_1 + S_2 + S_4 + S_6 + S_7 + S_8)/6$$

Similar corrections are to be made for all sensor and cylinder deviation calculations.

5.2 Re-adjustment of Reference Curve

If a sensor has been changed or relocated one of several possible re-adjustment procedures is to be performed by the operator in order to ensure good prediction of slow wear.

Three procedures are to be available for the operator to perform:

1. Re-adjustment of reference curve
 - is to be performed if a sensor has been changed or relocated
2. Re-adjustment of reference curve - and extrapolation of slow wear
 - is to be performed if a sensor has been out of operation for more than 100 operating hours
3. Re-adjustment of reference curve - and reset of slow wear
 - is to be performed if repairs have been performed which render the slow wear values invalid

Parts of the three procedures are similar: Re-adjustment of the reference curve due to the change in distance between sensor and guide shoe.

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When a sensor is changed or relocated historical information about slow wear is to be included in the re-adjustment of the reference curve.

5.2.1 Re-adjustment of Reference Curve

The procedure consists of a part a) of 50 hours duration and a part b).

- a) During 50 hours of offset re-adjustment, the signal from the sensor in question is to be compensated according to the existing rpm-dependent reference curve of this sensor.

While normally giving values distributed around zero, the compensated values of the sensor will now reflect the physical offset of the sensor.

The compensated values are to be averaged over 50 hours of operation, after which the reference curve of the affected sensor is offset accordingly, i. e. to indicate zero wear.

During the 50 hour re-adjustment alarms or slowdown requests due to slow wear related to the affected sensor are suspended. Rapid wear alarms and slowdown requests are to remain active.

- b) An offset (O_1) of the reference curve is to be performed to account for previous slow wear.

The offset is to be based on the stored 6 hour averages of compensated and filtered sensor values. The last 50 running hours of useful data are to be averaged (approximately 8 lines in the stored file) and the reference curve offset accordingly. See figure 5.1.

5.2.2 Re-adjustment of Reference Curve – and Extrapolation of Slow Wear

The procedure consists of a part a) of 50 hours duration, a part b) and a part c).

- a) During 50 hours of offset re-adjustment, the signal from the sensor in question is to be compensated according to the existing rpm-dependent reference curve of this sensor.

While normally giving values distributed around zero, the compensated values of the sensor will now reflect the physical offset of the sensor.

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The compensated values are to be averaged over 50 hours of operation, after which the reference curve of the affected sensor is offset accordingly, i. e. to indicate zero wear.

During the 50 hour re-adjustment alarms or slowdown requests due to slow wear related to the affected sensor are suspended. Rapid wear alarms and slowdown requests are to remain active.

- b) An offset (O_1) of the reference curve is to be performed to account for previous slow wear.

The offset is to be based on the stored 6 hour averages of compensated and filtered sensor values. The last 50 running hours of useful data are to be averaged (approximately 8 lines in the stored file) and the reference curve offset accordingly. See figure 5.1.

- c) An offset (O_2) of the reference curve is to be performed to account for slow wear occurring during time of breakage (t_b).

O_2 is an approximate value based on the slow wear recorded by the other sensor in the same unit (cylinder neighbour) during the time of breakage. A best fit Root Mean Square (RMS) line based on the 6 hours average of compensated and filtered sensor values of the cylinder neighbour during time of breakage is to be made. O_2 is the difference between the values at the two end points of this line. See figure 5.2.

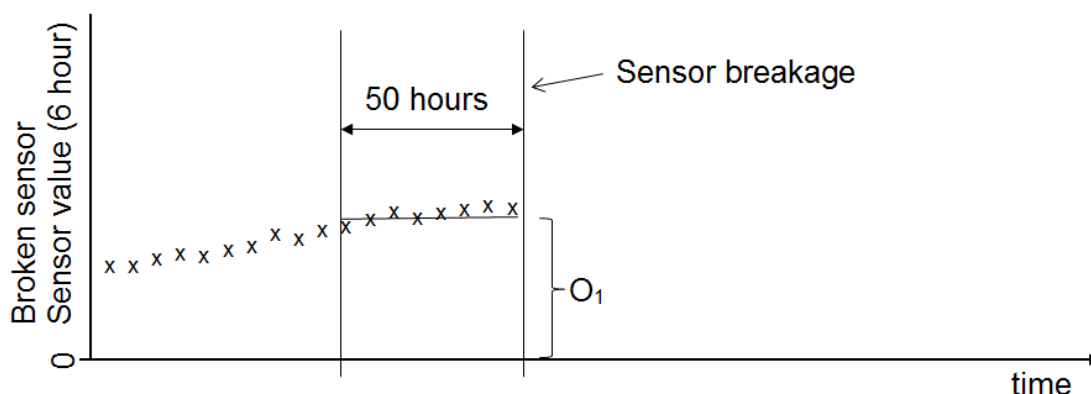


Figure 5.1, Calculation of sensor offset O_1 .

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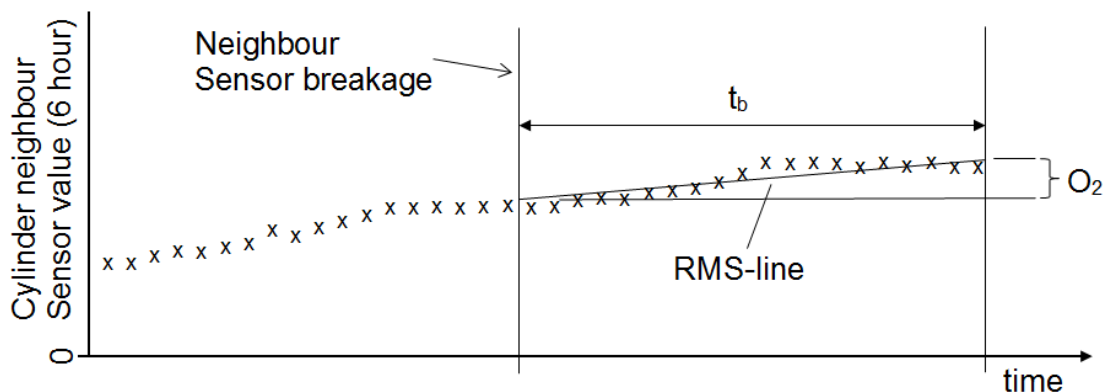


Figure 5.2, Calculation of approximated wear (O_2) during time of breakage.

5.2.3 Re-adjustment of Reference Curve – and Reset of Slow Wear

The procedure consists of one part a) of 50 hours duration.

- a) During 50 hours of offset re-adjustment, the signal from the sensor in question is to be compensated according to the existing rpm-dependent reference curve of this sensor.

While normally giving values distributed around zero, the compensated values of the sensor will now reflect the physical offset of the sensor.

The compensated values are to be averaged over 50 hours of operation, after which the reference curve of the affected sensor is offset accordingly, i. e. to indicate zero wear.

During the 50 hour re-adjustment alarms or slowdown requests due to slow wear related to the affected sensor are suspended. Rapid wear alarms and slowdown requests are to remain active.

5.2.4 Further Actions in Case of Sensor Adjustments

If any of the abovementioned readjustment procedures are performed it is to be noted in the log file.

If a sensor is adjusted the Rapid Wear sensor value for the adjusted sensor is reset according to:

Rapid Fast = 5 Rev Speed Compensated Average

Rapid Slow = 0

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6 Storage and Presentation of Data

Data storage in the BWM system serves three purposes:

- One purpose is to be able to retrieve relevant data in case of a bearing damage in order to investigate the physical phenomena leading to the damage.
- The second purpose relates to the inspection of bearings. Traditionally, bearings are opened up for routine inspections on a time-based schedule. E.g. every bearing must be opened at, say, 4-year intervals. To avoid unnecessary openings of the bearings, it is desired to shift from time-based inspections to condition-based inspections. For this purpose, trend curves showing if any wear has taken place during the inspection interval, are to be presented to the class surveyor.
- Thirdly, sufficient data for investigating the causes of false alarms and for debugging the software and calculation algorithms is to be available.

6.1 Frozen Copies

In the following sections several references to frozen copies of data are given. The frozen copies are to be made when alarms are released. Alarms may be released several times within a relatively short period of time if the crew decides to continue operating the engine even though an alarm has been released. This could lead to several frozen copies with overlapping data. Such overlapping frozen copies can be condensed to one frozen copy covering the full period.

Frozen copies are to be stored for at least one year. If storage space limits the amount of data which can be stored data can be erased earlier. In that case frozen hit-by-hit data covering 65 hours of operation should always be available for download. This corresponds to 5 individual alarm events.

All stored data are to be kept in a non-volatile memory.

6.2 Data to Store for Bearing Damage Analysis

There is no specification of the format for storage of data. But it must be possible to extract the data to text files with plain text of a specified format.

The BWM system must facilitate storage of the following data:

- Hit-by-hit data of uncompensated and unfiltered signals with time stamp and engine rpm for the latest 12 hours of operation. If an alarm limit is exceeded, a copy of the hit by hit storage is to be taken and stored separately as a "frozen" copy. **In case of a**

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rapid wear alarm the copy is to be extended to cover the time until the alarm has ceased and at least 1 hour of operation more. In case of a slow wear alarm the copy is to be extended to cover the time until the alarm was released and at least 1 hour of operation more. (See Appendix E for the required format of the unloaded data. There is one line in the file per revolution of the engine)

- Every time the speed compensation curves are changed the values of the curves are to be stored in a separate time stamped file. (See Appendix F for the required format of the unloaded data)

The files mentioned above give sufficient information to recreate any event leading to an alarm.

6.3 Data to Store for Debugging Purposes

The following data is to be available for the BWM-system manufacturer and MAN Diesel & Turbo to be able to study the BWM-system responses to events leading to alarms. The data is to make it possible to debug software and algorithms to assure that the system performs as intended.

- Rapid Wear storage for every revolution of the latest 2 hours of running that can be downloaded in case of an alarm or slowdown. If an alarm limit is exceeded, a copy of the short term storage is to be taken and stored separately as a "frozen" copy. The copy is to be extended to cover the time until the alarm has ceased and at least 5 minutes of operation more. As a minimum the following data is to be stored: timestamp, rpm, Rapid wear sensor, Rapid fast, Rapid slow, SingleRapid, CylRapidAve, TwinCylAve, deltaRpm, Rapid Alarm. No format for this storage is specified.
- Short term storage is to contain the speed compensated and filtered values, with time stamp and engine rpm, available from the latest 24 hours of operation. A minimum density of one set of data for every 30 engine revolutions is required. If an alarm limit is exceeded, a copy of the short term storage is to be taken and stored separately as a "frozen" copy. The copy is to be extended to cover the time until the alarm has ceased and at least 5 minutes of operation more. No format for this storage is specified.

6.4 Data to Store for Wear Documentation

In order to evaluate the wear history of the bearings the following data is to be stored:

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- Long term storage is to contain maximum, minimum, and average filtered values, for every 6 running hours, with time stamp. This storage must extend at least 5 years back. No format for this storage is specified. But it must be possible to extract the data to a text file with plain text.
- An event log has to contain the following information:
 - Any alarms, slowdowns or pre-warnings released or changes during the learning process of the BWM system.
 - Possible replacement and/or offset adjustment of any sensors.
 - Change of sensor reference from rough compensation to fine compensation
 - Possible reset of reference level for pre-warning
 - All information must be time stamped.

Downloading of stored data and event log to an external device like a PC must be possible for the purpose of presentation to surveyors from classification societies. For this purpose one XML file has to be made, including the four parts seen below. Their format is defined in appendix A-D.

In the files, the sensors are to be named by cylinder number (1=for) and then 1 (fore) or 2 (aft) as an example 7.1 and 7.2 for cylinder 7.

- Engine info part.** This is to contain information of the ship and engine. See Appendix A
- Log part** of all changes in system status and sensor status according to the event log. All the changes are to be presented with time stamp. See Appendix B
- Filtered value trend part.** Derived from “Long time storage”. Following data to be presented for each sensor every six engine operating hours: Time stamp, Engine operating hours, and 6 h average filtered value. See Appendix C.
- Status part.** The purpose of this file is to provide a quick overview of each cylinder showing if opening of the bearings related to this cylinder for inspection is justified due to detected wear or due to other circumstances. These circumstances could be detection of wear exceeding the “significance limit” during exchange of a sensor or loss of reference (see section 4). The status is to be shown as a four level presentation. The four levels are to be:
 - ❖ Normal “N” indicating that no wear has been detected beyond the pre-warning limit.
 - ❖ Pre-Warning “W” indicating that wear has been detected beyond the pre-warning limit.

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- ❖ Alarm "A" indicating that an alarm has been released from sensors connected to this cylinder.
- ❖ Unknown "U" indicating that a sensor of this cylinder has lost reference or its reference curve has been corrected due to wear detected by its "neighbour" during exchange. The same indication is to be given if a damaged sensor remains unchanged.

The system must provide the possibility to reset the status to "N".

Each part of the XML-file has to have the possibility to be generated individually and then with a time stamp when the file is generated.

6.5 User request for data storage

The system must provide the user of the system the possibility to make frozen copies of the 12 hour hit-by-hit data, the 2 hour Rapid Wear storage and the 24 hour short term storage at any time.

At least the two latest user requested data storages are to be available for download. Earlier user requested data storages can be erased to save storage space.

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7 Alarm Limits

7.1 Slow Wear Alarm Limits

The expected rpm-change of the measured value depended on engine size is shown in table 1.

Engine type (cm Bore)	mm change (20-110% rpm)
Above 75	-0.35
55 to 75	-0.25
Below 55	-0.15

Table 7.1, Engine type characteristic slope

If the processed signals exceed the below limits, an alarm or slowdown request is to be released:

Alarm limit, filtered sensor value	+/- 0.5 mm
Alarm limit, sensor deviation	+/- 0.4 mm
Alarm limit. cylinder deviation	+/- 0.3 mm
Slowdown limit, filtered sensor value	+/- 0.7 mm
Slowdown limit, sensor deviation	+/- 0.5 mm

Table 7.2, Alarm limits for slow wear algorithms during normal running

During learning conditions the limits are higher:

Alarm limit, filtered sensor value	+/- 0.8 mm
Alarm limit, sensor deviation	+/- 0.5mm
Alarm limit. cylinder deviation	+/- 0.4 mm
Slowdown limit, filtered sensor value	+/- 0.9 mm
Slowdown limit, sensor deviation	+/- 0.7 mm

Table 7.3, Alarm limits for slow wear algorithms during learning

The limit to get a pre-warning is:

Pre-warning limit, compensated sensor value (6hr. avg.)	+/- 0.25 mm
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Table 7.4, Pre-warning level

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7.2 Rapid Wear Alarm Limits

Alarm _{Basic,SingleRapid}	200 µm
Alarm _{Basic,CylRapidAve}	90 µm
Alarm _{Basic,TwinCylRapidAve}	75 µm

Table 7.5, Alarm limits for rapid wear algorithms during normal running.

Alarm _{Basic,SingleRapid}	230 µm
Alarm _{Basic,CylRapidAve}	120 µm
Alarm _{Basic,TwinCylRapidAve}	105 µm

Table 7.6, Alarm limits for rapid wear algorithms during refining of compensation curve.

Alarm _{Basic,SingleRapid}	260 µm
Alarm _{Basic,CylRapidAve}	150 µm
Alarm _{Basic,TwinCylRapidAve}	135 µm

Table 7.7, Alarm limits for rapid wear algorithms during creation of rough compensation curve.

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8 Interface to Alarm and Monitoring System

A stand-alone Bearing Wear Monitoring system must include the following connections (tag numbers refer to MDT Guidance Values):

Alarm relay, normally closed (XS8126): Must be triggered at pre-warning and alarm levels. When a pre-warning or alarm is active and a different alarm level is reached, the alarm relay must be toggled (closed-opened-closed) for a short period to trigger the alarm system again. Toggle period should be 1 second as default and adjustable in the range 0.1 to 5 seconds (minimum).

Slowdown relay, normally open with terminals for supervision resistor (XS8127): Must be triggered at slowdown levels.

System Fail relay, normally closed (XS8129): Must be triggered at power fail, watch dog fail or similar problem causing the system monitoring to be unavailable or unreliable.

For integrated systems, the same functionality must be proven, but hardware terminals may not be available.

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Appendix A, Engine Info Part

Name of Ship	= XXXXXXXX
IMO Number	= XXXXXXXX
Class Register No.	= XXXXXXXX
Component	= XXXXXXXX
Engine licenser	= XXXXXXXX
Engine maker	= XXXXXXXX
Engine type	= XXXXXXXX
Engine serial no.	= XXXXXXXXX
CM System Type	= Bearing Wear Monitoring System
CM System Maker	= XXXXXXXXX
CM System Hardware	= XXXXXXXX
CM System Software	= XXXXXXXX
Period of Trend Data From	= YYYY-MM-DD
Period of Trend Data to	= YYYY-MM-DD
Engine Operating hrs. From	= 99999
Engine Operating hrs. To	= 99999

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Appendix B, Log Part

2008-01-25, 10:43:21, Speed-compensation started
 2008-01-25, 10:43:21, Sensor No. 11.2, Speed compensation alarm
 2008-01-25, 10:43:21, Speed-compensation finished
 2008-02-11, 02:18:07, Sensor No. 3.1, Unknown
 2008-02-12, 12:11:17, Sensor No. 3.1, Re-adjustment started
 2008-02-13, 05:22:13, Sensor No. 6.2, Warning, Sensor Value
 2008-02-15, 16:08:59, Compartment No. 06, Warning
 2008-02-17, 02:18:07, Cylinder No. 6, Warning, Cylinder Deviation
 2008-02-19, 08:37:12, Sensor No. 6.2, Alarm, Sensor Deviation
 2008-02-23, 18:27:33, Compartment No. 06, Alarm
 2008-02-28, 11:53:23, Sensor No. 6.2, Slowdown release, Sensor Deviation
 2008-03-04, 12:11:17, Sensor No. 3.1, Re-adjustment alarm
 2008-03-08, 05:06:44, Sensor No. 3.1, Re-adjustment finished
 2008-03-14, 17:00:00, System, Change of alarm limit cylinder deviation from 0.3mm to 0.35mm by Administrator



Info. No.: 341402		Item Name: Bearing Wear Monitoring				Item Id.: 3091686-8		
Scale:	Size: A4	Product Type: Specification of Signal Processing and Operational Limits for Engines Coupled to a Fixed Pitch Propeller				Projection: 	Page No.: 35 (38)	
Date	Des.	Chk.	A.C.	Revision Change			Chg. Not.	Rev.
20180918	NIHE	HWC	X4	Small modifications to rapid wear during learning				07

Appendix C, Filtered Sensor Value Trend Part

Sensor No. 6-2

Date and Time [YYYY-MM-DD hh:mm:ss];Engine operating hours [Oh];Distance [mm]

2007-01-19 02:13:00;30000;0.144

2007-01-19 10:13:00;30006;0.133

2007-01-19 16:13:00;30012;0.151

2007-01-19 22:13:00;30018;0.213

2007-01-20 06:13:00;30024;0.218

2007-01-21 08:13:00;30030;0.133

2007-01-21 14:13:00;30036;0.161

Info. No.: 341402		Item Name: Bearing Wear Monitoring			Item Id.: 3091686-8		
Scale:	Size: A4	Product Type: Specification of Signal Processing and Operational Limits for Engines Coupled to a Fixed Pitch Propeller			Projection: 	Page No.: 36 (38)	
Date	Des.	Chk.	A.C.	Revision Change		Chg. Not.	Rev.
20180918	NIHE	HWC	X4	Small modifications to rapid wear during learning			07

Appendix D, Status File

2007-01-21 08:13:00

Legend: N=Normal; U=Unknown; W=Warning; A=Alarm

Cylinder No. 01	N
Cylinder No. 02	N
Cylinder No. 03	N
Cylinder No. 04	W
Cylinder No. 05	W
Cylinder No. 06	W
Cylinder No. 07	N
Cylinder No. 08	N
Cylinder No. 09	N
Cylinder No. 10	N
Cylinder No. 11	U
Cylinder No. 12	N

Info. No.: 341402		Item Name: Bearing Wear Monitoring			Item Id.: 3091686-8	
Scale:	Size: A4	Product Type: Specification of Signal Processing and Operational Limits for Engines Coupled to a Fixed Pitch Propeller			Projection: 	Page No.: 37 (38)
Date	Des.	Chk.	A.C.	Revision Change		Chg. Not.
20180918	NIHE	HWC	X4	Small modifications to rapid wear during learning		07

Appendix E, Hit-by-hit Data of Uncompensated Sensor Values

At every revolution hit-by-hit data of uncompensated and unfiltered signals with time stamp, engine rpm and engine direction are recorded. The direction is given as 1: forward or -1: reverse. Direction: 0 can be stated if the data for some reason is invalid.

The format is stated below:

Date[YYYY-MM-DD];Time[hh:mm:ss];Speed[rpm];Direction[-];S1.1[μm];S1.2[μm];S2.1[μm];S2.2[μm];S3.1[μm];S3.2[μm];S4.1[μm];S4.2[μm];S5.1[μm];S5.2[μm];S6.1[μm];S6.2[μm];S7.1[μm];S7.2[μm];S8.1[μm];S8.2[μm];S9.1[μm];S9.2[μm];S10.1[μm];S10.2[μm];S11.1[μm];S11.2[μm];S12.1[μm];S12.2[μm]

An example for a 5 cylinder engine is given below. The first line is a header line:

```
date;time;rpm;dir;S1.1;S1.2;S2.1;S2.2;S3.1;S3.2;S4.1;S4.2;S5.1;S5.2;
2016-08-30;22:42:52;88;1;59;27;212;-26;64;-169;-16;-166;157;-26;
2016-08-30;22:42:53;88;1;49;34;233;-49;71;-173;-23;-162;174;-48;
2016-08-30;22:42:54;88;1;48;28;229;-49;70;-180;-32;-169;181;-48;
2016-08-30;22:42:54;89;1;66;14;256;-64;39;-140;3;-192;212;-71;
...
```

More data can be added at the end of each line if the BWM-system provider wishes to.

Info. No.: 341402		Item Name: Bearing Wear Monitoring			Item Id.: 3091686-8	
Scale:	Size: A4	Product Type: Specification of Signal Processing and Operational Limits for Engines Coupled to a Fixed Pitch Propeller			Projection: 	Page No.: 38 (38)
Date	Des.	Chk.	A.C.	Revision Change		Chg. Not.
20180918	NIHE	HWC	X4	Small modifications to rapid wear during learning		07

Appendix F, Storage of Compensation Curves

Every time the speed compensation curves are changed the values of the curves are to be stored in a separate time stamped file.

If for instance the compensation curves were changed 2016-08-30 at 22:42:52 the data could be stored in a file named: compensation_curves_20160830_224252.txt

The format is stated below:

Start_Speed_interval[rpm];End_Speed_interval[rpm];S1.1[μm];S1.2[μm];S2.1[μm];S2.2[μm];S3.1[μm];S3.2[μm];S4.1[μm];S4.2[μm];S5.1[μm];S5.2[μm];S6.1[μm];S6.2[μm];S7.1[μm];S7.2[μm];S8.1[μm];S8.2[μm];S9.1[μm];S9.2[μm];S10.1[μm];S10.2[μm];S11.1[μm];S11.2[μm];S12.1[μm];S12.2[μm]

An example for a 5 cylinder engine with a nominal speed of 99 rpm is given below.
The first line is a header line:

Rpm1;Rpm2; S1.1;S1.2;S2.1;S2.2;S3.1;S3.2;S4.1;S4.2;S5.1;S5.2;
19.8; 20.6; -47; -81; -32; -97; -45; -71; -135; -10; -353; -29;
20.6; 21.4; -43 -77; -30; -96; -43; -68; -133; -9; -350; -28;
21.4; 22.2; -39 -73; -28; -95; -41; -65; -130; -8; -347; -27;
22.2; 23.0; -35 -70; -25; -95; -39; -62; -128; -7; -345; -27;
23.0; 23.8; -31 -66; -23; -94; -37; -59; -125; -6; -342; -26;

...
... 103 lines of data missing in the example...

...
106.2;107.0; 225;224;183;57;126;212;129;80; -98;115;
107.0;107.8; 235;233;197;72;136;193;123;97; -96;113;
107.8;108.6; 245;241;211;87;147;174;118;114; -95;111;
108.6;109.4; 255;250;225;102;157;155;112;131; -93;109;